

NIL Youth Sports Platform — Execution Plan (v2)

Document version: 2.0 — supersedes v1 build order **Date:** 2026-09-30 **Authority:** see §0.2 **Companion:** SPEC_AMENDMENTS.md (the patch list that makes the source spec set consistent with this plan) **Predecessor analysis:** SPINOFF_MVP_ANALYSIS.md (findings F1–F12 referenced throughout)

0. How to Use This Plan

0.1 Scope

Dimension	Value
MVP boundary	Stages 0–9 (Implementation Phases 1–3), delivered API-first
Stage 10	Post-MVP (Phase 4) — outline only, no estimates
Build order	Hybrid: extract API + schema (S1–S3), build and <i>validate</i> a vertical slice (S4–S5), then complete web extraction (S6) and feature breadth (S7–S9)
Relationship to source platform	Pattern replication — the spinoff reuses the source architecture, including the FlexibleApiSingleton hierarchy
Estimate	39.5–73 engineer-weeks, likely ~56 (§6)

0.2 Authority & supersession

Document	Status after this plan
PROJECT_SEQUENCE.md	Superseded as build order. Retained as historical record and as the stage → source traceability map (§9)
IMPLEMENTATION_PLAN.md	Superseded as task backlog. Its skill index (§0.1) and Definition of Done (§0) are retained and incorporated into §2.4 and §7
TECHNICAL_SPEC.md	Remains authoritative for architecture, DDL, capability design, and the §12 gap analysis — <i>as amended by SPEC_AMENDMENTS.md</i>
MIGRATION_DESIGN.md	Remains authoritative for the extraction inventory (KEEP/DROP lists, file counts) — <i>as amended</i>
FRONTEND_SPEC.md	Remains authoritative for UI surfaces,

Document	Status after this plan
	components, and navigation — <i>as amended</i>
SPINOFF_MVP_ANALYSIS.md	Findings register. Every finding F1–F12 is closed by a task or a gate in this plan (§8)

Conflict rule. If any document contradicts this plan, **this plan wins**. If this plan is silent, SPEC_AMENDMENTS.md is authoritative over the original spec text. If both are silent, the source platform's actual behavior wins over all documents (per finding F1 — this rule exists because the spec set asserted a mechanism that does not exist).

0.3 Work units

Unit	Definition
ew	engineer-week = 1 person × 5 focused days, inclusive of design, implementation, review, and the task's acceptance gate
Task	Smallest unit that can be independently verified. Has an ID, dependencies, deliverable, acceptance, effort, and confidence
Stage	A group of tasks with a single entry gate and exit gate
Gate	A hard stop. Cannot be passed on "mostly working." Requires the named evidence artifact

0.4 Confidence ratings

Rating	Meaning	Planning use
High	Comparable work has been done in this codebase; scope is enumerable	Use the estimate
Medium	Scope is enumerable but execution has unknowns	Use the estimate, add 20% buffer
Low	Scope depends on coupling that has not been measured	Do not commit. Measure first (spike, S0.1) or re-baseline after S5

Low-confidence stages in this plan: **S2, S3, S6, S9**. Together they are ~44 ew of the ~56 ew midpoint. That is the honest shape of the risk.

1. Decisions Ledger

Every decision below has a **recommended default**. The plan is executable immediately by accepting defaults; each may be overridden before its "Decide by" stage. Decisions marked **[SIGN-OFF]** require a human (legal, finance, or product owner) — they are not the implementer's to make.

ID	Decision	Recommended default	Rationale	Blocks	Decide by	Owner
D1	Is RLS the Phase-2 isolation guarantee or Phase-4 hardening?	Phase-4 hardening. Phase-2 enforcement = explicit WHERE tenant_id + a repository-level tenant guard (S4.1)	The source platform has no RLS (F1). Promising it in Phase 2 means promising a control that does not exist. The tenant guard gives equivalent bug-catch power at a fraction of the cost	S2, S4	S0	Eng lead
D2	Athlete-tenants: widen tenants or side table? What tier do they carry?	Add tenant_type + a nil_tenant_profile side table. Seed a platform_default tier row; force directory_visible = false for tenant_type='athlete'	tenants is commerce-shaped (Stripe slots, SKU quotas, GBP tokens). Athletes must not inherit those columns or public directory eligibility (F2)	S2	S0	Eng lead
D3	Which base-singleton chain is canonical?	Universal Singleton → EnhancedF	The spec's §13 hierarchy targets this	S0 (hygiene), S3, S6	S0	Eng lead

ID	Decision	Recommended default	Rationale	Blocks	Decide by	Owner
	Are V2/Stable deleted?	FlexibleApiSingleton → FlexibleApiSingleton → {Public, Tenant, Customer, Admin, Authenticated} is canonical. Delete FlexibleApiSingletonV2, FlexibleApiSingletonStable, PublicApiSingletonStable	chain. Three coexisting chains is drift, not a feature (F5)			
D4	Rename orders/order_items/payments, or create nil_deals/nil_deal_milestones/nil_payments fresh?	Create fresh, delete commerce models, reuse the services	On a fresh DB the rename buys nothing and drags Stripe/refund/tax/shipment entanglement along (F7)	S2, S3	S0	Eng lead
D5	Can an athlete-tenant appear in tenant-directory / discovery / storefront-discovery surfaces?	No — ever. Explicit exclusion rule + directory_visible = false default + a regression test	directory_visible defaults true ; athlete-tenants would otherwise be publicly discoverable	S2, S4	S0	Eng lead + product

ID	Decision	Recommended default	Rationale	Blocks	Decide by	Owner
			(F2, F9)			
D6	Are <code>nil_guardian / nil_fan_network</code> capability-gated or platform-default?	Platform-default, not tier-gated (per §12.9). Remove them from the capability registry and delete the corresponding hooks from <code>FRONTEND_SPEC.md</code> §13a.11	§7 and §12.9 contradict each other; §12.9 is the correct policy (never paywall the trust/safety and engagement layer) (F6d)	S9	S4	Product owner [SIGN-OFF]
D7	How does right-to-erasure coexist with an append-only audit trail?	Audit rows carry pseudonyms references only — never PII. Erasure rewrites the reference, not the log	Both requirements are legally load-bearing and the spec never reconciles them. Resolving it at schema time is cheap; later it is a migration (F9)	S2	S0	Eng lead + legal [SIGN-OFF]
D8	Test strategy: harness, CI gates, jsdom-or-not for web?	vitest everywhere. CI gates: typecheck, raw-fetch, UUID/Date.now() id, PII-under-/api/public/*. Web = server-render assertions	"Release-blocking P0 tests" must not be built after the features they guard (F10)	S0, S4	S0	Eng lead

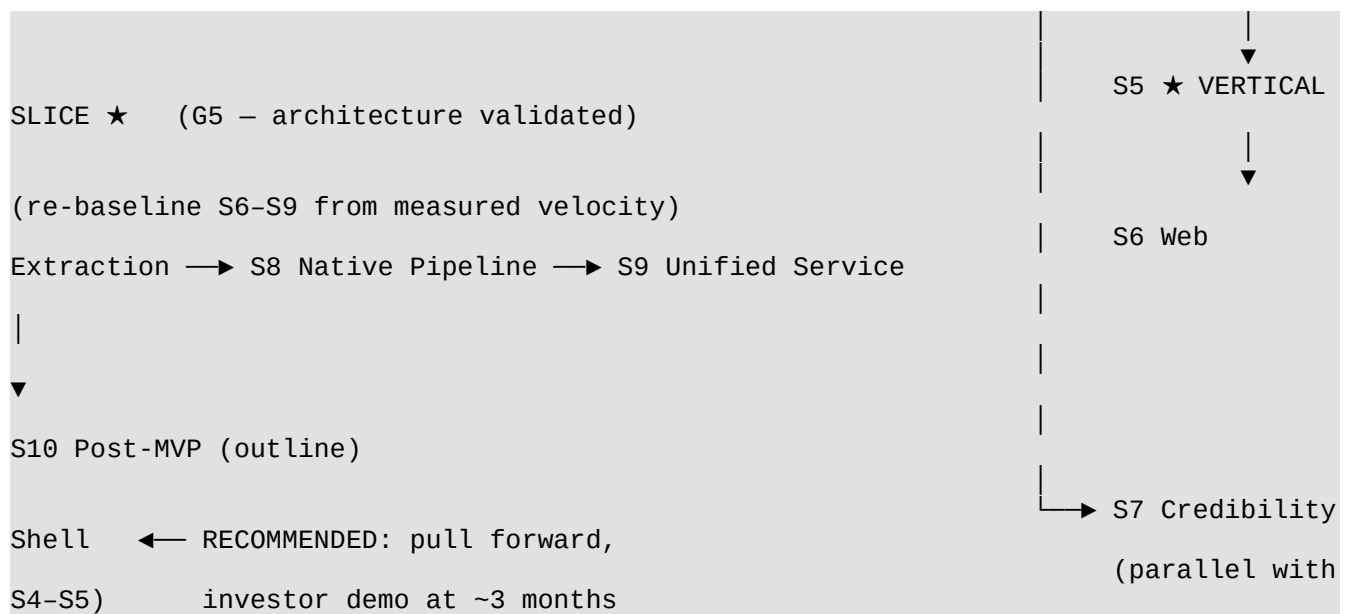
ID	Decision	Recommended default	Rationale	Blocks	Decide by	Owner
		only for now (matches existing AGENTS.md constraint)				
D9	Fee parameters: platform %, guardian split, non-profit slice, payer-keyed tier matrix	Defer. Model is already settled (§12.10 — deal = purchase, transaction fee). These are parameters	Confirmed non-blocking: parameter tuning, not architecture. Must be closed before S9 finance work, not before S1	S9 only	S8	Finance [SIGN-OFF]
D10	Legal sign-off on COPPA / FERPA / state-NIL / erasure design	Start immediately . External dependency, longest lead time	Hard-gates the schema freeze (S2). Slipping this slips everything	S2	S0 (start)	Legal [SIGN-OFF]
D11	Is a pilot dataset to be carried over?	Assume no — new Supabase project is empty	The whole plan assumes no data migration. If this is wrong, every estimate is void	S2	S0	Product owner

Rule: a stage does not begin while a decision it depends on is open. If a decision is not made by its "Decide by" stage, the recommended default is **adopted in writing** and the plan proceeds — no stage blocks on silence.

2. Delivery Model

2.1 Stage flow

S0 Baseline → S1 Infra → S2 Schema → S3 API Extraction → S4 Safety Substrate



Why this order (the hybrid choice). Worth being precise about the gain rather than overselling it.

- **Architectural feedback arrives at a similar time.** G5 lands at ~25 ew cumulative (S0–S5 mid). Under the spec's order, the equivalent validation lands at ~28 ew (A–D plus E, before Phase 2 begins). The feedback is *not* dramatically earlier — anyone claiming a large timing win is misreading the plan.
- **What genuinely improves is what you are holding when the feedback arrives.** This plan validates **before** the low-confidence destructive web strip (S6, ~9 ew) and never breaks the web app — it stays demoable from S1 onward. The spec's order strips the web app to zero *before* anything has been proven. If the architecture is wrong here, you correct a working app and a rebuilt API; under the spec's order you correct a half-destroyed one.
- **A third benefit, easily missed:** the API is built **once**, against a design that has been validated, rather than speculatively and then reworked.

Sequencing option — pull the Credibility Shell forward (recommended). S7 (landing + lead capture + nil_landing) has **no dependency on the slice**. It needs only S1 (infra), S2 (the nil_leads_list table), and S3.5/S3.7 (resolver plumbing + the lead service). Because Implementation Phase 1 is explicitly investor-facing, **S7 should be executed immediately after S3, in parallel with S4–S5.**

- Effect: an investor-ready public shell at **~21 ew cumulative instead of ~36.5 ew** — roughly 3 months instead of 8.
- Trade-off: nil_landing (tier-only, no merchant prefs) becomes the template-setting capability rather than nil_roster. A simpler reference is acceptable, and arguably preferable as a first pass.
- It does **not** disturb the G5 gate, the P0 suite, or the post-G5 re-baseline rule.

Serialization reality. $S2 \rightarrow S3 \rightarrow S4 \rightarrow S5$ is a hard chain and cannot be widened. $S7$ can move (above). $S6$ and $S9$ parallelize well. Plan staffing around the chain, not around the total.

2.2 Additive-then-subtractive rule (*optimization O4*)

Never delete a Prisma model before the code that references it is gone. The spec's M1/M2 split deletes models and code in the same pass, which means every deletion batch produces a cascade of type errors from both directions at once.

Mandated order:

1. **Add** all NIL models (additive — breaks nothing). `prisma validate` green.
2. **Delete** commerce code in vertical batches, `checkapi` green after each batch.
3. **Then** drop commerce models, `grep` proving zero references remain.

This converts one large entangled repair into two ordered, independently-verifiable repairs.

2.3 Definition of Done

Per task (inherited from `IMPLEMENTATION_PLAN.md §0`, retained):

- `pnpm checkapi` and `pnpm checkweb` pass with **zero TS errors** (verified baseline — both green today)
- The task's named acceptance check passes
- No raw `fetch` introduced (`web` → `makeDefaultRequest`; `API` → `UniversalSingleton` helpers)
- No `randomUUID()` / `Date.now()` IDs — all IDs from `id-generator.ts`
- **Exception (documented):** `bot_*` tables retain `@db.Uuid + gen_random_uuid()` per `TECHNICAL_SPEC.md §14.12`. The `grep` gate must whitelist `apps/api/src/services/bot/` and the `bot` route directory, or the gate fails on compliant code (F6c)

Per stage: the exit gate's evidence artifact exists and is attached to the stage PR.

Per capability (the 8-phase pipeline, from `TECHNICAL_SPEC.md §7` — corrected paths per F4):

1. **Define** feature keys + tier assignment in the seed scripts (`apps/api/prisma/seed-*` - `capabilities.ts`, `seed-tiers.ts`) — **not** `canonical-features.ts` / `tier-hierarchies.ts`, which do not exist
2. **Seed** `features_list` → `capability_features_list` → `tier_features_list`
3. **Store prefs** `tenant_nil_*_options_settings` model + table
4. **Resolve** `resolveNilXxx()` + wire `EffectiveCapabilityResolver.ts` + add the disabled entry to `buildExpiredCapabilitiesResponse`

5. **Route** `nil-xxx-options-settings.ts` (GET { success, settings, tierState }; PUT tier-validates + invalidateEffectiveCapabilities)
6. **Map** `UnifiedCapabilityService.mapNilXxx()` + state interface + `useNilXxxCapability` hook
7. **Display** `PlanSummaryPanel` + `CapabilityShowcase` (correct group-level `merchantGated` counting)
8. **Verify** `checkapi` + `checkweb` + `verify-capability-deployment` checklist

2.4 Standing engineering rules

Rule	Enforcement
No raw <code>fetch</code> in NIL code paths	CI grep gate
No <code>randomUUID/Date.now()</code> IDs (bot exception above)	CI grep gate
No PII fields on <code>/api/public/*</code>	CI grep gate + S4.6 enumeration script
Concrete NIL services extend a NIL base , never <code>UniversalSingleton</code> directly	CI check (F5 — prose rules demonstrably did not hold in the source repo)
Every minor-data mutation enumerates all cache namespaces	<code>cross-context-cache-invalidation.md</code> + S4.4 contract + test
Every status/consent/deal/payout transition emits an <code>X-Audit-ID</code>	<code>structured-logging.md</code> + S4.8
Before building any <code>nil_*</code> capability, open its commerce analog and copy the structure	Task-level checklist item (the spec's own §7.1 rule, made explicit)

3. Gates

Each gate has an **evidence artifact** — a file, test run, or report attached to the stage. "Passed" without the artifact is not passed.

Gate	After	Exit criteria	Evidence artifact
G0	S0	Spike report with measured repair ratio; green baseline captured; CI gates running; D1–D5, D7, D8, D11 closed; D10 legal review formally started; live-DB object inventory complete	<code>spike-report.md</code> , <code>baseline-typecheck.txt</code> , CI config committed, <code>db-object-inventory.md</code>

Gate	After	Exit criteria	Evidence artifact
G1	S1	Unmodified fork deploys to Railway + Vercel; Auth0 login → dashboard → log out round-trips for all 8 roles; doppler secrets lists every required key in all 3 configs; zero old-brand strings	deploy logs, Auth0 role screenshot set, grep output
G2	S2	prisma validate green; NIL wave-1 tables created on the new DB; seeds run idempotently; API still compiles (additive only — no deletions yet); tenant_type + nil_tenant_profile + platform_default tier present; directory_visible=false for athlete type	prisma validate output, seed logs, checkapi output
G3	S3	checkapi green; zero commerce route/service references; commerce Prisma models dropped with grep proof of no references; 19 ID generators with tests; resolver registry dispatches; 8 NIL roles validate	checkapi output, grep proofs, generator test run
G4	S4	Tenant guard rejects a query against an athlete-owned table with no tenant predicate (proven by test); consent engine handles grant/revoke/most-restrictive-wins; visibility state machine enforces all transition	test run, CI log, pii-surface-report.md

Gate	After	Exit criteria	Evidence artifact
		guards; cache contract enumerated; PII enumeration script runs in CI; negative-path harness executes	
G5 ★	S5	Architecture validated. Slice demo: guardian provisions athlete-tenant → grants scoped consent → passes moderation + compliance → profile appears on public roster → revoke removes it within one request. All P0 negative tests green. PII gate green	demo recording, P0 test run, PII report
G6	S6	checkweb green; zero commerce web references; 7 NIL bases + web services in place on the canonical chain; proxy/auth/capability-service/rbac/nav updated; rebrand complete (no VisibleShelf strings)	checkweb output, grep proof, brand audit
G7	S7	Landing live on the NIL domain over TLS; lead intake writes and is not publicly readable; nil_landing capability resolves + R13 expired-manifest returns 200 disabled	TLS check, lead write test, capability route test
G8	S8	M2 milestone: media/metrics/achievements live; institution roster management; manual compliance verdict workflow; every P0 negative-path test from §12.12 green	full P0 suite run
G9	S9	All personas have	capability checklists ×5,

Gate	After	Exit criteria	Evidence artifact
		scoped portals; 5 persona capabilities deployed (8 phases each); CRM + bot live with child-safety guardrails; invitations + onboarding working end-to-end; every adult ↔ minor interaction guardian-gated	P1 suite run, invitation E2E

★ G5 is the decision gate. After G5, re-baseline S6–S9 from measured velocity (§6.5). Do not commit to the S6–S9 numbers before then.

4. Stages

Stage 0 — Baseline, Hygiene & Decisions

Entry: none. Exit: G0. Estimate: 1.5–2.5 ew.

ID	Task	Depends	Deliverable	Acceptance	ew	Conf
0.1	Extraction spike (<i>de-risking</i>): on a throwaway branch, delete one complete commerce vertical (inventory/recommended), repair until checkapi and checkweb are green. Record files deleted, files <i>edited</i> , errors surfaced,	—	spike-report.md with measured repair ratio per vertical	Ratio extrapolates to the full DROP list; two lowest-confidence stage estimates replaced with measured numbers	0.6–1.0	High

ID	Task	Depends	Deliverable	Acceptance	ew	Conf
	wall-clock					
0.2	Capture green baseline: checkapi + checkweb outputs; record file/LOC/model counts	—	baseline-typecheck.txt, baseline-metrics.txt	Artifacts committed; baseline reproducible	0.1	High
0.3	Install CI gates: typecheck, raw-fetch, UUID/Date.now() id, PII-under-/api/public/*, prisma validate, "NIL service must extend a NIL base"	0.2	CI config committed	Each gate fails on a deliberately-introduced violation (gate-testing the gate)	0.3–0.5	High
0.4	Hygiene purge: 509 .bak/.backup files, 3 backup Prisma models (subscription_tiers_list_v1_backup, tier_features_list_v1_backup, tenants_metadata_backup_gbp), abandoned singleton variants per	D3	Clean tree	grep -cE "\.(bak\ backup)\$" → 0; prisma validate green; checkapi/checkweb still green	0.5–1.0	High

ID	Task	Depends	Deliverable	Acceptance	ew	Conf
	D3, stray baseline dumps					
0.5	Live-DB object inventory: enumerate triggers, policies, views, materialized views, extensions from the live Supabase DB — not from <code>schema.prisma</code>	—	db-object-inventory.md	Inventory reconciles against <code>schema.prisma</code> ; every out-of-band object listed with a keep/drop decision	0.2–0.4	High
0.6	Close the decisions ledger (D1–D5, D7, D8, D11) or adopt defaults in writing	—	Signed-off ledger	Every blocking decision has a written answer or an adopted default	0.1	High
0.7	Start legal review (D10) — external, longest lead time	—	Legal engagement confirmed + target date	Review formally opened and tracked as an S2 dependency	—	—

Why 0.1 before everything. The estimate's uncertainty is concentrated in the deletion-and-repair tax. A 3–5 day measurement replaces the two lowest-confidence rows with measured numbers and typically halves the total range. It also produces the coupling map (which retained subsystems touch dropped models) that S2 needs anyway.

Why 0.4 is not optional. 509 backup files and three backup Prisma models make every downstream grep gate noisy, and leave duplicate-looking models a junior agent can mistake for the real one. It is 0.5–1.0 ew that improves every subsequent verification step.

Stage 1 — Infrastructure

Entry: G0. Exit: G1. Estimate: 1–2 ew. (= spec Stage A)

ID	Task	Deliverable	Acceptance	ew	Conf
1.1	New git repo; clone source files without .git history	Clean repo, single initial commit	git log shows one commit	0.1	High
1.2	New Supabase project; CREATE EXTENSION vector	Provisioned project	Dashboard accessible; extension present	0.2	High
1.3	New Auth0 tenant + 8 roles (athlete, guardian, institution_ admin, coach, sponsor, fan, platform_ admin, compliance_ auditor); callback + logout URLs	Configured tenant	All 8 roles visible; login round-trips locally	0.3–0.5	Medium
1.4	New Vercel project (web)	Linked project	Preview deploy succeeds	0.1	High
1.5	New Railway service (api)	Linked service	Service deploys	0.1	High
1.6	Doppler: local, dev, prd configs with all secrets	Three configs	doppler secrets lists every required key in each	0.2–0.4	Medium
1.7	Package rename (@rvp/* → @nil/*); brand strings; PLATFORM_D OMAINS; API_BASE_U	Renamed packages	pnpm install succeeds; zero old-brand references	0.2–0.4	High

ID	Task	Deliverable	Acceptance	ew	Conf
	RL defaults				
1.8	Prove the unmodified fork deploys to both targets	Deploy logs	Both targets serve; API health check passes	0.1–0.2	High

Task 1.8 is the real point of this stage. Deploying *before* any deletion separates "the fork is broken" from "our extraction broke it." Without it, the first deploy failure in S3 is ambiguous.

Stage 2 — Schema Foundation

Entry: G1. **Exit:** G2. **Estimate:** 4–8 ew. (= *spec Stage B, re-scoped: additive only*)

This stage adds. It does not delete. Per §2.2, commerce model removal happens in S3.6, after the code that references those models is gone.

ID	Task	Depends	Deliverable	Acceptance	ew	Conf
2.1	Wave-1 schema (slice-critical): enums; tenants.tenant_type; nil_tenant_profile; athlete_profiles_list; guardians_list; guardian_athlete_links_list; consent_records_list; highlight_media_li	D2	Migration nil_wave1	prisma validate green; tables created; athlete provisions as a tenant	1.0–2.0	Low

ID	Task	Depends	Deliverable	Acceptance	ew	Conf
	st; nil_leads _list; the *_options _settings base					
2.2	Wave-2 schema (breadth): athlete_m etrics_li st, athlete_a chievemen ts_list, athlete_t enant_mem berships_ list, recruitin g_boards_ list, scout_rat ings_list , sponsorsh ip_deals_ list, escrow_mi lestones_ list, nonprofit _allocati on_pools_ list, nil_eligi bility_ru les_list, moderatio n_cases_l ist, message_t hreads_li st, fan_badge s_list,	2.1, D4	Migration nil_wave2	prisma validate green; table count reconciles against TECHNICAL _SPEC.md §14 counted directly (the plan's list undercounts — F12)	1.5–3.0	Low

ID	Task	Depends	Deliverable	Acceptance	ew	Conf
	data_eras ure_reque sts_list, payout_sc hedules_l ist, sponsor_s pend_limi ts_list, nil_offer s_list, nil_event s_list, nil_invit ations_li st, nil_onboa rding_ses sions_lis t, 7 × tenant_ni l*_optio ns_settin gs					
2.3	platform_ default tier + NIL feature/capab ility/tier seed scripts	D2, D6	Seed scripts	Athlete/ guardian/fan features resolve from platform default, never tier_feat ures_list	0.5–1.0	Medium
2.4	Seed base data: NIL eligibility rules, navigation links, bot guardrails (child-safety)	2.2	Seed scripts	All base data present; seeds idempotent	0.5–1.0	Medium
2.5	mv_athlet e_discove ry materialized	2.2, D5	View + refresh function + scoping note	View queryable; proven to exclude	0.3–0.6	Medium

ID	Task	Depends	Deliverable	Acceptance	ew	Conf
	view + its scoping review			unapproved and directory_visible=false rows (F9)		
2.6	Guardian-required trigger (nil_require_guardian_for_minor) on CRM tables	2.2	Trigger	Minor-subject CRM record without guardian → DB error	0.2–0.4	High
2.7	Audit-trail schema per D7 (pseudonymous references; erasure reconciliation)	D7	Schema + design note	Erasure can rewrite references without falsifying the log	0.2–0.4	Medium
2.8	RLS decision applied (D1): if Phase-4 hardening, write the deferral record; if Phase-2, build the policies and the transaction-mode GUC workaround	D1	rls-decision.md (+ policies if option 2)	Decision recorded with rationale; if built, isolation test passes non-vacuously (proves rows are visible with the right tenant set)	0.2–0.4 / +3–6	Medium

F1 closure note. MIGRATION_DESIGN.md §3.4 currently states RLS "carries over unchanged." It does not exist. If you take the default (D1 = Phase-4 hardening), 2.8 is a short design record and the Phase-2 guarantee is the S4.1 tenant guard. If you take option 2, budget it here honestly — ~24 tables × (policy + GUC plumbing + Supabase transaction-mode workaround + non-vacuous isolation tests) is its own workload.

F12 closure note. Count the tables from TECHNICAL_SPEC.md §14 directly. The plan lists 22; the migration design lists 25; the spec contains 23 CREATE TABLE statements plus the options tables. Reconcile before writing the migration.

Stage 3 — API Extraction & Core

Entry: G2. Exit: G3. Estimate: 6–12 ew. (= spec Stage C)

ID	Task	Depends	Deliverable	Acceptance	ew	Conf
3.1	Delete commerce routes in vertical batches (products, inventory, storefronts, shops, directory, GBP, GMC, barcode, cart, shipments, business hours, feed, store reviews, featured, social commerce, recommendations, image enrichment, clone, quick-start, publishing, digital downloads, catalog adoption, cross-tenant products, location availability, deposit forfeiture, smart-sale	0.1 (spike ratio), 0.3	Deleted routes	checkapi green after each batch; no broken mounts in index.ts	2.0–4.0	Low

ID	Task	Depends	Deliverable	Acceptance	ew	Conf
	tagging, slug gen, override analytics)					
3.2	Delete commerce services + middleware (image-search-limits.ts, sku-limits.ts) + commerce-specific test/doc files	3.1	Deleted services	checkapi green	1.5–3.0	Low
3.3	19 NIL ID generators + table-driven tests	3.2	id-generator.ts entries + tests	All formats match §6; collision check done; test covers every generator	0.3–0.6	High
3.4	NIL resolver registry: NilLandingResolver, NilRosterResolver, NilGuardianResolver, NilRecruitingResolver, NilSponsorshipResolver, NilAchievementsResolver, NilFanNetworkResolver,	3.3	Resolver files	Each returns the correct capability shape. Build the first one by copying its commerce analog, then template the rest (O2)	1.0–2.0	Medium

ID	Task	Depends	Deliverable	Acceptance	ew	Conf
	NilComplianceResolver, NilFinanceResolver, NilCrmOptionsResolver, NilBotOptionsResolver					
3.5	Rewire Effective CapabilityResolver.ts: NIL imports, MerchantSettingsBundle type, dispatch, buildExpiredCapabilitiesResponse disabled entries	3.4	Updated orchestrator	resolveEffectiveCapabilities() dispatches to NIL resolvers; R13 expired manifest returns 200 disabled	0.3–0.6	Medium
3.6	Drop commerce Prisma models (second wave, per §2.2) — only after grep proves zero references	3.1, 3.2	Reduced schema.prisma	grep proof of zero references; prisma validate green; checkapi green	0.5–1.0	Medium
3.7	Phase-1/2 NIL backend services: NilLeadService, AthleteProfileServ	3.5	Services	Non-consent_authority write → 403; non-allowlisted host rejected	1.5–3.0	Medium

ID	Task	Depends	Deliverable	Acceptance	ew	Conf
	ice, GuardianC onsentSer vice, MediaMode rationSer vice, NilRoster Service, Complianc eVettingS ervice (manual verdict)					
3.8	Auth middleware + 8 NIL roles (auth.ts, role- validatio n.ts, permissio ns.ts)	1.3	Updated middleware	All 8 roles validate correctly	0.3–0.6	Medium
3.9	Route mounting in index.ts: remove commerce mounts, add NIL mounts	3.1, 3.7	Updated entry	Server starts without errors	0.2–0.4	High

O2 — template-first capability (*the single largest effort saver*). Task 3.4 builds eleven resolvers. Do **not** design them independently. Build `NilRosterResolver` by copying its commerce analog end to end, get it through the 8-phase pipeline (S5.5), then produce the remaining ten by analogy against that reference. Eleven independent designs is the failure mode; one reference plus ten analogies is roughly half the cost.

Stage 4 — Safety Substrate

Entry: G3. **Exit:** G4. **Estimate:** 2–4 ew. (*new stage — F1, F9, F10*)

This stage exists because the spec treats its safety controls as *features* of Phase 2 rather than as an **enforcement layer that Phase 2 features are built on**. Building the layer first means

every subsequent feature inherits it, instead of each feature re-implementing (and subtly mis-implementing) tenant scoping, consent checks, and cache eviction.

ID	Task	Depends	Deliverable	Acceptance	ew	Conf
4.1	Repository-level tenant guard (D1): a Prisma middleware/wrapper that rejects any query against an athlete-owned table without a tenant predicate	D1	Guard module	Test: a query against athlete_profiles_list with no tenant_id is rejected at runtime. This is the Phase-2 substitute for RLS (F1)	0.5–1.0	Medium
4.2	Consent engine: versioned scoped consent; consent_authority checks; most-restrictive-wins across multiple guardians; revocation cascade contract	2.1	GuardianConsentService core + tests	Grant/revoke/veto all tested; any owning guardian can veto; cascade contract enumerated	0.5–1.0	Medium
4.3	Visibility firewall state machine + transition guards (draft→pending→approved→archived, pending→rejected)	2.1	State machine module	Transition to approved requires all three gates true; illegal transitions rejected	0.3–0.6	Medium
4.4	Cache-invalidation	4.2, 4.3	Contract + test	Approve → roster reflects	0.3–0.6	Medium

ID	Task	Depends	Deliverable	Acceptance	ew	Conf
	contract: namespace enumeration + getServiceCachePatterns / invalidateServiceCaches + eviction test			within one request; revoke evicts every namespace (partial eviction = safety incident)		
4.5	Media moderation core + host allowlist (parseVideoUrl pattern)	2.1	Service + allowlist	Non-allowlisted host rejected; media cannot publish unmoderated	0.2–0.4	High
4.6	PII / public-surface enumeration script (<i>optimization — automated, not a manual audit</i>): walks all retained read paths (routes, materialized views, nav data, email templates, Sentry config, analytics, sitemap/preview) and reports any athlete/guardian field reachable	2.5	pii-surface-report.md + CI gate	Script runs in CI; report enumerates every surface; no unclassified surface remains (F9)	0.4–0.8	Medium
4.7	Negative-path test	D8	Harness	The five P0 tests from	0.3–0.6	Medium

ID	Task	Depends	Deliverable	Acceptance	ew	Conf
	harness (DB-level + API-level) for the P0 suite			§12.12 can be written and executed against it (F10)		
4.8	Audit trail implementation per D7: X-Audit-ID emission on every status/consent/deal/payout transition	D7, 2.7	Logging integration	Every transition emits an audit id; audit rows contain no PII	0.2–0.4	Medium

F9 closure. The spec's PII strategy covers DTO projections. Task 4.6 covers the paths a DTO test cannot reach: the materialized view, the directory default, DB-driven nav, email templates, error reporting, analytics, and sitemap/preview surfaces.

F10 closure. Task 4.7 is scheduled **before** the features it guards, not after.

Stage 5 — Vertical Slice Validation ★

Entry: G4. **Exit:** G5 (the decision gate). **Estimate:** 2.5–4.5 ew. (= the spec's "First Executable Slice," expanded to include its web consumer)

ID	Task	Depends	Deliverable	Acceptance	ew	Conf
5.1	Athlete-tenant provisioning (guardian-initiated) + COPPA age-band gate	4.1, 4.3	Provisioning path	Athlete-initiated under-13 intake → 403	0.4–0.7	Medium
5.2	Guardian link + scoped consent grant/revoke through the private 0-TTL path	4.2	Consent flow	Non-consent_authority write → 403; scoped grants enforced	0.3–0.5	Medium
5.3	Visibility	4.3, 5.1, 5.2	Working	Only all-	0.3–0.5	Medium

ID	Task	Depends	Deliverable	Acceptance	ew	Conf
	state machine end-to-end on a real profile		transitions	three-gates-true reaches approved		
5.4	Public roster route: approved-only, consent-filtered, PII-stripped	5.3	NilRoster Service + route	Zero records when no row is approved; GPA visible only with gpa_display consent	0.4–0.7	Medium
5.5	nil_roster + nil_compliance capabilities, full 8 phases — the reference implementation (O2)	3.4, 3.5, 5.4	Capability end-to-end	verify-capability-deployment checklist passes; R13 expired manifest works	0.6–1.0	Medium
5.6	Per-athlete cache eviction on status/consent change	4.4	Eviction wiring	Approve → roster reflects within one request; revoke → gone within one request	0.2–0.4	Medium
5.7	P0 gates: under-13 self-register 403; consent revocation cascade; media moderation gate	4.5, 4.7	P0 tests green	All three negative paths pass	0.3–0.6	Medium
5.8	Web slice (additive only): NilPublic ApiSingle ton,	D3, 5.4	Web slice	Slice renders against the live API; no commerce code deleted yet (hybrid	0.7–1.2	Medium

ID	Task	Depends	Deliverable	Acceptance	ew	Conf
	AthleteApiSingleton, GuardianApiSingleton, NilPublicRosterService, AthleteService, GuardianService + a minimal public roster page			property)		

★ **G5 is where the architecture gets validated.** If athlete-as-tenant, the tenant guard, scoped consent, the firewall, capability gating, and the P0 gates all hold on this thin path, the remaining ~40 ew is execution rather than discovery. If any of them does not hold, you have spent ~10 ew instead of ~26 to find out.

Task 5.8 keeps the hybrid property. The web app is *additive* here — the slice's consumers are new files. Commerce web code is still present and still compiling. Destructive web extraction is S6.

Stage 6 — Web Extraction & Core

Entry: G5 + re-baseline (§6.5). **Exit:** G6. **Estimate:** 6–12 ew. (= *spec Stage D*)

ID	Task	Depends	Deliverable	Acceptance	ew	Conf
6.1	Delete commerce web routes in batches (/products/*, /shops/*, /directory/*, /items/*, /cart/*, / carts/*, /	G5, 0.1 ratio	Deleted routes	checkweb green after each batch	2.0–4.0	Low

ID	Task	Depends	Deliverable	Acceptance	ew	Conf
	checkout/ *, /orders/* , /my- orders/*, /downloads/*, /catalog/ *, /category- discovery /*, /cross- tenant/*, /test-*, /debug/*, /sentry- example- page/*)					
6.2	Delete commerce web services, components, hooks, API route handlers	6.1	Clean src	checkweb green	1.5–3.0	Low
6.3	Remaining NIL bases + web services on the canonical chain (D3): ComplianceApiSingleton, InstitutionApiSingleton, SponsorApiSingleton, FanApiSingleton + their services	D3	Base classes + services	Each extends the correct NIL base; CI base-check passes	0.8–1.5	Medium

ID	Task	Depends	Deliverable	Acceptance	ew	Conf
6.4	proxy.ts + PLATFORM_DOMAINS; AuthContext / ServerResolvedContextProvider; UnifiedCapabilityService; rbac.ts	6.3	Updated infra	Auth flow works against the new Auth0 tenant; capability requests use NIL keys	0.5–1.0	Medium
6.5	Navigation links reseeded for NIL sidebars	6.4	Reseeded nav	Every sidebar item correct per FRONTEND_SPEC.md §8.3	0.3–0.6	Medium
6.6	Rebrand: colors, logo, fonts, email templates, Sentry project	6.4	Rebranded UI	Zero VisibleShelf branding remains	0.5–1.0	Medium

Stage 7 — Credibility Shell

Entry: G6. **Exit:** G7. **Estimate:** 2–3 ew. (= spec Stage E / Phase 1)

ID	Task	Depends	Deliverable	Acceptance	ew	Conf
7.1	Marketing landing route + NIL brand domain in proxy.ts env config	6.4	Landing segment	TLS handshake + 200 on landing	0.5–0.8	Medium
7.2	Lead intake form (athlete/parent vs sponsor	3.7, 7.1	Form + route + Zod schema	Lead writes succeed; no PII publicly readable	0.5–0.8	Medium

ID	Task	Depends	Deliverable	Acceptance	ew	Conf
	vs investor) → NilLeadService → nil_leads_list					
7.3	nil_landing capability, full 8 phases (second capability — now templated from S5.5)	5.5, 7.2	Capability	verify-capability-deployment checklist passes	0.5–0.8	Medium
7.4	Investor-facing shell review: state coverage, responsive breakpoints, all	7.3	Reviewed shell	skill-frontend-ux-guardrails checklist passes	0.3–0.6	Medium

Optimization visible here. nil_landing is the second capability built. With S5.5 as the reference implementation, its cost drops from a template-establishing exercise to an analogy. Tasks 7.3 and the five S9.2 capabilities all benefit from the same template.

Stage 8 — Native Pipeline Breadth

Entry: G7. **Exit:** G8. **Estimate:** 2.5–5 ew. (= remainder of spec Stage F / Phase 2 beyond the slice)

ID	Task	Depends	Deliverable	Acceptance	ew	Conf
8.1	Media + metrics + achievements : models, services, submission paths	2.2	Services + routes	Media defaults pending; achievements feed the profile only when approved	0.6–1.2	Medium
8.2	athlete_tenant_mem	2.2	Service + routes	Transfer preserves	0.4–0.8	Medium

ID	Task	Depends	Deliverable	Acceptance	ew	Conf
	berships_ list + AthleteMe membershipS ervice (transfers)			history; cross-tenant row visible to both tenants		
8.3	Institution roster management + admin/compl iance roster controller	8.2, 5.5	Controller UI + routes	Status toggle triggers per- athlete cache eviction	0.5–1.0	Medium
8.4	Manual compliance verdict workflow (Complian ceVetting Service full)	3.7	Workflow	No profile reaches approved without a verdict	0.3–0.6	Medium
8.5	Remaining P0 gates: non- allowlisted video host rejected; unmoderated media never public	4.5, 8.1	P0 tests	Both negative paths pass	0.3–0.6	Medium
8.6	RLS follow- through per D1 (if option 2: implement the deferred policies; if option 1: write the Phase-4 handoff note)	2.8	Policies or handoff note	Non-vacuous isolation test (option 2) or documented deferral (option 1)	0.4–0.8 / +1– 2	Medium

M2 milestone: a guardian can create an athlete-tenant, grant scoped consent, pass moderation + compliance, and see the athlete on the public roster — **with every P0 negative-path test green.**

Stage 9 — Unified Service

Entry: G8. Exit: G9. Estimate: 12–20 ew. (= spec Stage G / Phase 3)

ID	Task	Depends	Deliverable	Acceptance	ew	Conf
9.1	athlete_tenant_memberships_list transfers + multi-guardian edge cases	8.2	Service hardening	Most-restrictive-wins verified with conflicting guardians	0.4–0.8	Medium
9.2	Five persona capabilities, 8 phases each (D6): nil_recruiting, nil_sponsorship, nil_achievements, nil_compliance (auto-ready), nil_finance (per D9). nil_guardian / nil_fan_network are platform-default, not gated	5.5 (template), D6	Capabilities	verify-capability-deployment passes per capability; template-then-analogy, not independent design	3.0–5.0	Low
9.3	Six per-actor dashboards (guardian, athlete, institution, sponsor, fan, compliance) reusing CrmPageShell/CrmNavPanel +	6.4, 9.2	Dashboards	No polling loop; unread via getReadState; scoped per base; server-resolved context	3.0–6.0	Low

ID	Task	Depends	Deliverable	Acceptance	ew	Conf
	the 5 standard panels (FRONTEND_SPEC.md §3a)					
9.4	CRM deltas: athlete_tenant_id/guardian_id columns + guardian-required trigger + tenant_nil_crm_options_settings; NIL CRM surfaces (Compliance / Institution+Sponsor / Guardian); nil_crm capability	2.6	Services + routes + capability	Minor-subject ticket without guardian → DB error; PII never projected without a consented relationship	1.0–2.0	Medium
9.5	Bot deltas: conversation athlete/guardian scoping + is_minor_safe + tenant_nil_bot_options_settings; 4 personas (Compliance/Eligibility, Guardian Onboarding, Recruiting/Sponsor FAQ, Fan);	2.2	Bot configs + RAG + capability	No minor PII in responses; unanswered → CRM escalation; guardrails at the RAG retrieval filter, not just the prompt	1.0–2.0	Medium

ID	Task	Depends	Deliverable	Acceptance	ew	Conf
	bot_guard rail_rule s seed; nil_bot capability					
9.6	Invitations: InvitationService (lifecycle, guardian consent gate, anti-spam) + 9 routes + 5 FE components + 6 email templates + 6 connection-establishment side effects (each with its cache-invalidation obligation)	9.2	Invitation subsystem	Create/accept/reject/withdraw/guardian-approve work; minor cannot self-accept ; 30-day re-invitation block enforced; each connection type creates the correct row + fires invalidation	2.0–3.5	Low
9.7	Onboarding: OnboardingService (actor-aware state machine) + 3 routes + OnboardingWizard + OnboardingProgress Banner	9.6	Onboarding subsystem	Per-actor step sequences return correct completion %; skip works for non-blocking steps	1.0–2.0	Low
9.8	Cross-cutting sweep: observability, cache correctness, i18n for compliance/b	9.2	Sweep report	Every standing rule in §2.4 verified	0.5–1.0	Medium

ID	Task	Depends	Deliverable	Acceptance	ew	Conf
	ylaw copy					

9.6 is the sleeper. Invitations and onboarding are described in the spec as one Phase-3 sub-feature. They are a bidirectional subsystem with nine routes, six side effects, six email templates, and a consent gate. Budget them as their own workstream.

Stage 10 — Post-MVP (outline only)

No estimates. Listed so nothing is silently dropped. Full scoping is a separate exercise after G9.

- Automated compliance vetting (`ComplianceVettingService` auto + `nil_eligibility_rules_list` expansion per state/association)
- Financial infrastructure: deal-as-purchase transaction fee, `EscrowLedgerService` double-entry, guardian KYC/W-9, payout routing, non-profit pool slice
- RLS implementation (if D1 = option 1) across all athlete-owned + cross-tenant tables
- `DataErasureService` cascade + certificate
- Age-out job (18 → athlete gains financial scope, consent ledger preserved)
- Advanced abuse detection / scraping defense / rate-limit tuning

5. Optimizations Applied

Each is a concrete change from the spec's approach, with the saving it produces.

#	Optimization	Replaces	Why it saves
O1	API-first hybrid sequencing (S1–S3 → S4–S5 → S6)	Full extraction before any feature work	Surfaces architectural error at ew ~10 instead of ew ~26. Keeps a compiling web app throughout, so the destructive web pass is isolated
O2	Template-first capability: build <code>nil_roster</code> end-to-end as the reference, then produce the other 10 by analogy	11 independently-designed capabilities	One design decision plus ten analogies instead of eleven designs. Largest single saving in S9
O3	Automated PII/public-	One-time manual audit	Repeatable, covers non-

#	Optimization	Replaces	Why it saves
	surface enumeration (S4.6) as a repeatable CI gate	+ DTO snapshot test	route surfaces, and catches regressions after G5
O4	Additive-then-subtractive schema sequencing (§2.2)	Spec's M1/M2 split (delete models and code together)	Two ordered, independently-verifiable repairs instead of one entangled cascade
O5	Tenant guard instead of Phase-2 RLS (S4.1, default D1)	24 RLS policies + GUC plumbing + Supabase transaction-mode workaround in Phase 2	Equivalent bug-catch power at ~1 ew instead of ~3–6 ew plus an unsolved pooling problem
O6	Single canonical base chain; delete V2/Stable (S0.4, D3)	Three coexisting chains carried into the fork	Removes drift and prevents new NIL bases from targeting an abandoned chain
O7	Table-driven ID generator tests (S3.3)	Per-generator ad-hoc verification	19 generators verified by one parameterized test
O8	CI gates from commit 1 (S0.3), each gate-tested against a deliberate violation	Adding gates later	The source repo's type-level health eroded precisely because the deploy build masks errors (F3). Gates added late never get adopted
O9	Hygiene purge before extraction (S0.4)	Cleaning up at the end	509 backup files make every grep gate and every "did we get them all?" verification noisy
O10	Spike-measured repair ratio (S0.1)	Guessing the deletion tax	Converts the two lowest-confidence estimates into measured ones
O11	Re-baseline S6–S9 after G5 (§6.5)	Committing to the full estimate up front	~44 ew of the estimate sits in Low-confidence stages; velocity after the slice is real data
O12	Deferred nil_finance/auto-compliance to S10	Pulling Phase-4 finance into the MVP	Keeps the MVP about the safety core, not money movement. D9 (fee params) is explicitly non-blocking

#	Optimization	Replaces	Why it saves
O13	Evidence artifact per gate (§3)	"Stage complete" by assertion	Makes gate passage auditable and prevents silent scope erosion
O14	Decisions ledger with defaults (§1)	Open questions blocking stages	No stage blocks on silence; the default is adopted in writing and the plan proceeds
O15	Server-resolved context for all dashboards (S9.3)	Client polling	Prevents the ~1/sec request storm the spec itself flags (§12.11)
O16	Pull the Credibility Shell forward (§2.1): run S7 immediately after S3, in parallel with S4–S5	Shell at the end of the plan (~36.5 ew cumulative)	Implementation Phase 1 is explicitly investor-facing, and S7 has no dependency on the slice. Investor-ready public site at ~3 months instead of ~8 . Zero cost in total effort — sequencing only

6. Effort

6.1 Estimate by stage

Stage	Low	Mid	High	Confidence	Dominant cost
S0 Baseline, hygiene, decisions, spike	1.5	2.0	2.5	High	Spike + hygiene + CI gates
S1 Infrastructure	1.0	1.5	2.0	High	Auth0 8-role wiring; Doppler parity; proving the fork deploys
S2 Schema Foundation	4.0	6.0	8.0	Low	Schema surgery + seeds + the RLS decision
S3 API Extraction & Core	6.0	9.0	12.0	Low	The deletion-and-repair loop across 542 route files / 361

Stage	Low	Mid	High	Confidence	Dominant cost
					services
S4 Safety Substrate	2.0	3.0	4.0	Medium	Tenant guard, consent engine, harness, PII enumeration
S5 Vertical Slice Validation ★	2.5	3.5	4.5	Medium	The thin end-to-end path + 3 P0 gates + web slice
S6 Web Extraction & Core	6.0	9.0	12.0	Low	Same repair loop at 168,776 LOC / 436 pages / 303 services
S7 Credibility Shell	2.0	2.5	3.0	Medium	Landing + leads + second capability
S8 Native Pipeline Breadth	2.5	3.5	5.0	Medium	Media/ metrics/ achievements + roster mgmt + manual compliance
S9 Unified Service	12.0	16.0	20.0	Low	Breadth: 5 capabilities, 6 dashboards, CRM, bot, invitations, onboarding
Total (S0–S9)	39.5	≈56	73.0		

6.2 Correction to the v1 analysis

SPINOFF_MVP_ANALYSIS.md §5.3 reported a midpoint of **~48 ew**; the correct midpoint of its own ranges was **~52 ew** (extraction 26.25 + feature 25.5) — an arithmetic slip in that document's summary line.

This plan's midpoint is **~56 ew**, i.e. **+4 ew over the corrected v1 figure**. The increase is deliberate and traceable:

Change	Δ ew (mid)	Why
S0 (baseline, hygiene, decisions,	+2.0	Previously implicit in

Change	Δ ew (mid)	Why
spike) added as a stage		"extraction"; the spike and hygiene pass are real work
Safety substrate (S4) + slice (S5) + breadth (S8) replace the single "Stage F" line	+3.0	The v1 figure of 5–9 ew for the entire native pipeline was optimistic by omission — it did not carry the tenant guard, the negative-path harness, or the PII enumeration as explicit line items
Total	+5.0	Offset by O2/O4/O5 savings already reflected in the per-stage numbers

Reporting the increase explicitly is the point: the v1 number was low for reasons, and hiding that would repeat the spec's own error (F11).

6.3 Backend / web split

The backend is **not** half the work — it is roughly two thirds, because schema, API extraction, and the safety core dominate and are shared.

Slice	Stages included	Low	Mid	High
Backend / API — infra, schema, API extraction, safety substrate, slice (API), shell (API), native pipeline, unified service (API)	S0, S1, S2, S3, S4, S5 (API), S7 (API), S8, S9 (API)	26.0	37.5	49.0
Web / frontend — slice (web), web extraction, web bases/services, landing + roster UI, 6 dashboards, ~44 components, nav, rebrand	S5 (web), S6, S7 (web), S9 (UI)	13.5	19.0	24.0
Total		39.5	≈56	73.0

6.4 Calendar scenarios

Assumes decisions are made on time. $S2 \rightarrow S3 \rightarrow S4 \rightarrow S5$ is a hard chain; S6 and S9 parallelize.

Team	Calendar (mid, ~56 ew)	Notes
1 engineer	12–17 months	Fully serialized on the S2 → S5 chain
2 engineers (backend, web)	7–10 months	Web is idle during S2–S5 — consider starting S6 prep in parallel
3 engineers (backend, web, safety/QA)	5–8 months	Best fit. Safety/QA owns S4.6/S4.7 and the P0 suite from S2 onward
4 engineers (+ data/platform)	4.5–7 months	Diminishing returns; the chain does not widen

6.5 Re-baselining rule (O11)

After G5, re-estimate S6–S9 from measured velocity. S2, S3, S6 and S9 are Low-confidence and carry ~44 ew of the ~56 ew midpoint. By G5 you have measured:

- the actual repair ratio (S0.1 predicted it; S3 confirmed it)
- the real schema surgery cost (S2)
- capability cost for one full 8-phase implementation (S5.5)
- slice velocity per task

Feed those into S6–S9 and reissue §6.1. **Do not treat the numbers above as a commitment beyond G5.** Treating them as one is exactly the failure mode documented in F11.

6.6 Cost

Effort converts to money in the companion document **BUDGET .md**. Headline figures for the recommended delivery shape (**2 human engineers + 3 agent seats**), using the illustrative loaded rate of \$3,000/ew:

	Mid	Range
MVP build (labor + agents + infra + legal)	~\$168,000	\$128,000–224,000
— of which human engineering labor	~\$130,000	
— of which legal review (parameter — hold open until quoted)	\$30,000	\$15,000–60,000
Post-launch infrastructure run-rate	~\$150–600/month	

Three things in that document change how **this** plan should be executed:

1. **Budget agent capacity by usage, not seats.** Anthropic's published enterprise data puts observed Claude Code cost at \$150–250/developer/month with the seat fee only ~8–12% of the total. A \$20 seat can produce a \$250 bill. Use **\$200–400/seat-month all-in**.
2. **~34 of the 56 ew midpoint is agent-suitable; ~22 ew is human-owned.** BUDGET .md §4.1 maps this per stage. The human-owned portion is S4 (Safety Substrate) and S5 (Vertical Slice) plus the review tax — and it is the part that is **not compressible**.
3. **Optimization O2 (template-first capability) is the single largest source of agent leverage** — one reference implementation, then ten near-copies. It is worth doing well for that reason alone, not just for the direct time saving.

The S0.1 spike should also measure agent effectiveness. Have the agent perform the deletion-and-repair, and record agent-hours, human review-hours, and defects found. That replaces BUDGET .md parameters 6 (E, the effectiveness multiplier) and 7 (T, the review tax) with measured values — the same 3–5 days the plan already recommends, now paying off twice.

7. Verification & Evidence Standard

7.1 Test layers

Layer	Tool	Covers	Where
Typecheck	tsc --noEmit via pnpm checkapi / pnpm checkweb	Both apps, zero errors	CI, every PR
Unit	vitest	ID generators, pure helpers, mappers, resolvers	apps/api, apps/web
Integration (API)	vitest + test DB	Services, consent engine, firewall transitions, capability routes	apps/api
DB-level	vitest + test DB	Tenant guard rejection, guardian-required trigger, MV scoping	apps/api
P0 negative paths	vitest + harness (S4.7)	The five §12.12 P0 tests	Release-blocking
Component render	vitest + renderToStaticMa rkup	Web components (no jsdom — see D8 / AGENTS.md)	apps/web
Static gates	grep / AST script	Raw fetch, UUID ids, PII on	CI, every PR

Layer	Tool	Covers	Where
		/api/public/*, NIL-base compliance	

7.2 P0 release-blocking suite (from *TECHNICAL_SPEC.md §12.12*)

- ☐ Under-13 athlete cannot be registered without prior verifiable guardian consent → 403
- ☐ No messaging thread between an adult role and a minor can exist without a guardian participant (*structural invariant*)
- ☐ Consent revocation removes the profile **and all media** from public surfaces within one request and evicts **every** cache namespace
- ☐ Deal creation is blocked where `nil_eligibility_rules_list.deals_allowed = false`
- ☐ Media cannot publish without moderation clearance; non-allowlisted hosts rejected

A stage cannot ship with any of these red — **even if it is functionally working**.

7.3 Frontend verification constraint

apps/web has **no jsdom and no testing-library** (per *AGENTS.md*). Component tests server-render via `renderToStaticMarkup`; `useEffect` never runs; Radix/Accordion/Tabs panels unmount when inactive. Consequences the plan must live with:

- Assert on serialized markup (`value="..."` attributes, copy presence), not on interaction
- Use `initialTab / initialOutcome`-style props to reach state-gated panels without interaction
- Extract pure helpers and data-contract mappers into named exports and unit-test those instead of the component
- The *FRONTEND_SPEC.md §14* checklist items that are *interactional* or *layout-based* ("no page-level horizontal overflow", "mobile 320px / desktop 1440px intentionally designed") are **not expressible** in this harness. They require manual review at S7.4, recorded as a signed checklist — not a passing test

If the project wants automated interaction/layout testing, that is a **deliberate dependency decision** (D8), made once, not discovered per-component.

8. Gap Closure Matrix

Every finding from *SPINOFF_MVP_ANALYSIS.md* maps to the task or gate that closes it. No finding is left to chance.

Finding	Severity	Closed by	How
F1 RLS premise is false; spec points at commented-out dead code	Critical	D1, S2.8, S4.1, S8.6, SPEC_AMENDMENTS .md TS-1/TS-8/MD-4	Decision ledger resolves it; tenant guard substitutes in Phase 2; RLS deferred to S10 or budgeted honestly; source docs corrected
F2 No <code>tenant_type</code> ; <code>tenants</code> is commerce-shaped; <code>directory_visible</code> defaults true	Critical	D2, D5, S2.1, S2.5, TS-9	Side table + <code>platform_default</code> tier + forced <code>directory_visible=false</code> + MV scoping test
F3 Build masks TS errors; <code>checkapi</code> is the only real gate	High	S0.2, S0.3, O8	Green baseline captured; typecheck gate installed and gate-tested from commit 1
F4 Spec file references stale or non-existent	High	S2.3 (corrected paths), §2.3, SPEC_AMENDMENTS .md TS-4/TS-7/MD-6	Every path replaced with a verified one before it is used in a DoD
F5 Three coexisting base-singleton chains	High	D3, S0.4, S6.3, §2.4, TS-10/MD-2	Canonical chain chosen; abandoned variants deleted; NIL-base rule enforced in CI
F6 Five internal spec contradictions	High	§1 (D1, D3, D6), §2.3 (bot UUID exception), SPEC_AMENDMENTS .md	Each contradiction resolved to one authoritative answer; losing text amended
F7 <code>orders/payments</code> rename drags entanglement	Medium	D4, S2.2, S3.6, MD-3	Create <code>nil_deals/nil_deal_milestones/nil_payments</code> fresh; delete commerce models; reuse services
F8 509 backup files + backup Prisma models carried into the fork	Medium	S0.4, O9	Hygiene purge before extraction
F9 Non-route PII surfaces unswept (MV, directory, nav, email, Sentry, audit/erasure conflict)	Critical	D5, D7, S2.5, S2.7, S4.6, S4.8, O3	Automated enumeration script as a CI gate; MV scoping test; audit rows pseudonymous; erasure reconciliation designed at schema time

Finding	Severity	Closed by	How
F10 P0 tests have no harness; frontend harness constrained	Medium	D8, S4.7, §7.3	Harness built before the features it guards; web constraints documented and the interactional checklist routed to manual review
F11 Estimates exist only for extraction, and are ~an order of magnitude low	Low	§6 (this plan), O10, O11	Effort published for every stage; spike measures the tax; S6–S9 re-baselined after G5
F12 NIL model count undercounted	Low	S2.2	Tables counted directly from <code>TECHNICAL_SPEC.md</code> §14 before the migration is written

9. Traceability

Stage → source documents, so nothing is silently dropped from the original plan.

This plan	Spec stage	Source
S0 Baseline & hygiene	<i>(new)</i>	<code>SPINOFF_MVP_ANALYSIS.md</code> §5.6, §6 items 1/4/11; F1–F12
S1 Infrastructure	Stage A	<code>MIGRATION_DESIGN.md</code> §1 (M0); <code>PROJECT_SEQUENCE.md</code> A
S2 Schema Foundation	Stage B	<code>MIGRATION_DESIGN.md</code> §3 (M1); <code>PROJECT_SEQUENCE.md</code> B; <code>TECHNICAL_SPEC.md</code> §14, §18
S3 API Extraction & Core	Stage C	<code>MIGRATION_DESIGN.md</code> §4 (M2); <code>PROJECT_SEQUENCE.md</code> C; <code>TECHNICAL_SPEC.md</code> §6, §7, §9
S4 Safety Substrate	<i>(new — distributed in spec)</i>	<code>TECHNICAL_SPEC.md</code> §3, §8, §12.6–12.8, §12.12; <code>IMPLEMENTATION_PLAN.md</code> 2.13–2.15

This plan	Spec stage	Source
S5 Vertical Slice	(spec's "First Executable Slice")	PROJECT_SEQUENCE.md "First Executable Slice"; TECHNICAL_SPEC.md §8, §10
S6 Web Extraction & Core	Stage D	MIGRATION_DESIGN.md §4.7–4.9 (M3); PROJECT_SEQUENCE.md D
S7 Credibility Shell	Stage E / Phase 1	PROJECT_SEQUENCE.md E; TECHNICAL_SPEC.md §4 Phase 1
S8 Native Pipeline	Stage F / Phase 2	PROJECT_SEQUENCE.md F; TECHNICAL_SPEC.md §4 Phase 2, §12.12
S9 Unified Service	Stage G / Phase 3	PROJECT_SEQUENCE.md G; TECHNICAL_SPEC.md §4 Phase 3, §15–§18
S10 Post-MVP	Stage H / Phase 4	PROJECT_SEQUENCE.md H; TECHNICAL_SPEC.md §4 Phase 4

10. Risk Register (v2)

Ordered by expected impact. Each has a **trigger** — the observable signal that the risk is materializing — so mitigation starts before the damage.

#	Risk	Sev	Likelihood	Trigger	Mitigation	Contingency
R1	RLS assumed present, is absent → Phase-2 privacy guarantee weaker than documented; isolation tests pass vacuously	Critical	Certain (verified)	Any test asserting isolation that passes with zero rows visible	D1 + S4.1 tenant guard + non-vacuous test requirement (S2.8)	Build RLS in S2 (+3–6 ew) and slip S7
R2	Deletion-and-repair	High	High	S0.1 spike ratio > 3×	O10 spike; §2.2	Cut breadth: ship S1–S8,

#	Risk	Sev	Likelihood	Trigger	Mitigation	Contingency
	tax exceeds estimate			spec assumption; a batch takes $> 1.5\times$ its estimate	additive-then-subtractive; batch with <code>checkapi</code> per batch	defer S9 capabilities to a second release
R3	Minor data reaches a public surface via a non-route path	Critical	Medium	S4.6 enumeration reports an unclassified surface	S4.6 automated gate + S2.5 MV scoping test + D5 exclusion + D7 pseudonymous audit	Halt S6; treat as a release blocker
R4	<code>tenant_type</code> / athlete-tenant tier semantics left ambiguous	High	High	Capability resolver called with an athlete-tenant and returning a tier-shaped answer	D2 + S2.1 + S2.3 <code>platform_default</code> tier	Add an explicit actor-type guard to the resolver
R5	Spec instructions remain internally contradictory → wrong variant of a safety control implemented	High	High	Implementer asks "which document wins?"	§0.2 conflict rule + <code>SPEC_AMENDMENTS.md</code> + §1 decisions ledger	Freeze the spec set; re-issue amendments before continuing
R6	Consent revocation cascade is partial → stale minor data publicly cached	Critical	Medium	A revoke test passes but a <i>different</i> namespace still serves the profile	S4.4 enumerated contract + P0 cascade test (release-blocking)	Halt feature work; fix eviction completeness before proceeding
R7	Erasure vs. immutable audit trail unreconciled	High	Medium-High	Legal review flags it, or an erasure request cannot complete	D7 + S2.7 pseudonymous references designed at schema time	Retrofit = data migration; avoid by deciding in S0

#	Risk	Sev	Likelihood	Trigger	Mitigation	Contingency
R8	State NIL law changes mid-build	High	High (inherent)	A state passes/amends HS NIL legislation	Data-driven <code>nil_eligibility_rules_list</code> — no bylaw logic in code	Seed update only; no code change
R9	Frontend verification gap → UX guardrails silently skipped	Medium	High	Components merged with no state/layout evidence	D8 + §7.3 + S7.4 signed manual checklist	Add jsdom deliberately (a D8 revision)
R10	Bot/RAG leaks non-consented athlete data	Critical	Medium	A bot response contains a non-consented field	Guardrails at the RAG retrieval filter , not the prompt; <code>is_minor_safe</code> default true	Disable <code>nil_bot</code> capability until fixed
R11	Supabase pooler transaction mode breaks the tenant-context mechanism	Medium	Medium	GUC does not persist across a pooled query	Resolve in S2.8 before scheduling RLS	Use a direct connection for RLS-scoped work
R12	Fork inherits dead weight → noisy gates, duplicate models, incomplete sweep	Medium	Certain (verified)	Any grep gate returning <code>.bak</code> noise	S0.4 hygiene purge	—
R13	A pilot dataset exists after all	Medium	Low	Supabase project is not empty	D11 confirmed at S0	Void the estimate; re-plan with a data-migration stage
R14	Auth0 8-role mapping breaks after	Medium	Medium	A role fails to resolve on the new	S1.3 + S3.8 + G1 round-trip	Fix at S1 — do not carry a broken auth

#	Risk	Sev	Likelihood	Trigger	Mitigation	Contingency
	the fork			tenant		into S3

11. Change Control

Change type	Process
Task-level (reorder, split, re-estimate a task)	Update this document; note the reason in the stage PR
Gate change (exit criteria relaxed)	Requires the plan owner's explicit sign-off, recorded in this document. Relaxing a P0 criterion is not permitted without a written risk acceptance
Stage insertion/removal	Requires the plan owner's sign-off + a re-issued §6 estimate
Decision override (a D-number)	Update §1 with the override, the rationale, and the date. Overrides that contradict a <code>SPEC_AMENDMENTS.md</code> entry require that entry to be revised too
Finding closure disputed	Re-open the finding in <code>SPINOFF_MVP_ANALYSIS.md</code> , mark §8 here as contested, and re-verify against the repo before proceeding

Re-baseline checkpoint. §6.1 must be re-issued after G5 (§6.5). A stage that begins after G5 using the pre-G5 estimate without a re-baseline is a plan violation.

12. Summary

Question	Answer
Is the approach sound?	Yes. Pattern replication of a verified-capable platform, with athlete-as-tenant as the isolation root
Is the plan executable now?	Yes. Every blocking decision has a recommended default (§1); every finding has a closing task or gate (§8); every stage has an entry gate, exit gate, and evidence artifact (§3)
Build order	Hybrid (API-first): S1–S3 extract API + schema, S4–S5 validate a vertical slice, S6 extracts web, S7–S9 add breadth

Question	Answer
Estimated effort	39.5–73 ew, likely ~56. Backend ~37.5 ew (⅔), web ~19 ew
Calendar	3 engineers: 5–8 months . Solo: 12–17 months
When is the estimate trustworthy?	After G5. ~44 of ~56 ew sits in Low-confidence stages; re-baseline S6–S9 from measured velocity (§6.5)
The single highest-value action	S0.1 — the 3–5 day extraction spike. It converts the two lowest-confidence estimates into measured numbers and produces the coupling map S2 needs
The single highest-risk item	R1 / F1 — resolve the RLS premise (D1) before S2 begins. Do not carry the false premise into the fork
What must not slip	D10 legal review. External, longest lead time, hard-gates the S2 schema freeze